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Task Description:

The SHERE II experiment was operated on board the International Space Station (ISS) during Increment 27/28. SHERE II is an extension of the SHERE project, and both used the same hardware but different test fluid. The experiment is a glovebox investigation developed at NASA Glenn Research Center (GRC) and intended for operation in the Microgravity Science Glovebox (MSG) aboard the ISS. It is the first instrument capable of assessing the effects of a pre-shear history on the extensional properties of dilute polymer solutions during the extended range of time scales that can be accessed because of the absence of gravitational sagging.
As a logical extension to the SHERE project, SHERE II looks to study the effect of torsional pre-shear on the extensional behavior of a different non-Newtonian fluid. We specifically chose to study dilute suspensions of rigid micro-particles in dilute polymer solutions (0.025 wt. %). The suspending particles used are rigid inert poly(methyl methacyrlate) (PMMA) spheres, 6 µm in diameter, and the polymeric matrix is narrow polydispersity high molecular weight polystyrene dissolved in oligomeric styrene, which is the same fluid used in SHERE.

The SHERE II experiment will allow us to measure for the first time the effects of rotational pre-shear on the axial stretching of a polymeric fluid with small particles dispersed within it. The experiment also allows the study of the fluid's relaxation after the axial stretching, which is only possible in a microgravity environment. The results will provide new insights into the behavior of non-Newtonian fluids in industrial processes with both rotational and axial flows. The results of SHERE II will provide high quality experimental data to compare with recently developed theories on filler materials in plastics. Comparison of the SHERE (no particles) and SHERE II (particles) experiments is expected to be enlightening.

The goal of the SHERE II experiment is to study the effects of a rotational pre-shear history on the transient extensional viscosity in a uniaxial stretching flow for dilute suspensions of inert, rigid micro-spheres (particles) suspended in a viscoelastic matrix fluid consisting of a dilute polymer solution. Constitutive equations for viscoelastic fluids such as dilute polymer or wormlike micelles solutions generally predict large changes in the extensional rheology of the material as the elongation rate is increased. However, the validity of these theories has not yet been confirmed due to the lack of experimental data obtained in extensional flows. Prior to this research, quantitative rheological measurements in shear-free flows were only possible for highly elastic materials such as polymer melts which could easily be elongated without sagging under a gravitational body force. The SHERE and SHERE II experiments allow us to obtain, for the first time, accurate measurements of the extensional viscosity for less elastic fluids such as polymer solutions, suspensions and liquid crystalline materials.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Understanding the rheological properties of highly viscoelastic suspensions may be of paramount importance for lunar in-situ resource utilization and for the future construction of a permanent lunar base. Indeed, the lunar soil, referred to as regolith, is composed of loose dust and broken rocks that could be used as a high volume fraction filler when mixed with a polymeric binder or matrix for building foundation or roadbed materials. The viscoelastic dilute suspension used for SHERE II thus serves as a highly idealized model system amenable to theoretical analysis which can shed light on future development of lunar regolith-based construction materials.
Suspensions of particles in viscoelastic liquids are used in many Earth processing operations: polymer melts with fillers, ceramic pastes, biomedical materials, food, cosmetics and detergents. The final properties of the suspensions are greatly determined by the shape, concentration and size of the filler. In particular, the fillers can range from nanoscopic to microscopic characteristic dimensions, which leads to very different types of flow behaviors, filler/matrix interactions and dynamics.

The results of SHERE II will provide new insights into the behavior of this important class of non-Newtonian fluids in industrially relevant mixed shear and extensional flows. This information is particularly relevant for understanding the deformation and evolution of fluid threads in a wide variety of industries, including polymer processing, food and consumer product processing, biomedical diagnostics, and pressure-sensitive adhesives. This research has direct applications in many industrial processes such as fiber spinning, film coating, injection molding, extrusion, and blow molding. Additionally, shear free flows arise whenever liquid interfaces are present; for example in "containerless processing" operations. These rheological data obtained will enable designers of both space- and ground-based material processes to use improved constitutive models in numerical simulations of complex two- and three-dimensional fluid flows. This includes stability and breakup of jets, enhanced oil recovery, and turbulent drag reduction for advanced aircraft, boats, and submarines.

The main research task completed during this reporting period is the complete analysis of the data received upon completion of SHERE II experiments. These experiments were carried out in the Microgravity Science Glovebox (MSG) aboard the International Space Station (ISS), and the data was delivered to the PI's laboratory on 06/25/12. The analysis was performed using the high definition videos of the filament captured during experiments. The current progress in the analysis confirms the preliminary results obtained from the lower-quality down-linked videos, as well as presents many new findings.
The SHERE II experiment consists of three distinct regimes: the pre-shear regime, the stretch regime and the capillary thinning regime. MATLAB codes were written and optimized for each of these regimes. Analysis of the experimental results reveals that the difference in temperature between the start and end of the test is within specified requirements for a successful test.

We first examine the results of the stretch regime of the experiment. The displacement profile of the endplate provides an extension rate in agreement with the expected imposed value. Inspection of the high definition videos acquired during experiments shows that as the left-hand endplate commences to displace exponentially to the left, a symmetric fluid filament is formed and stretched homogeneously between the two endplates. Finally, when the stretching phase has completed the fluid drains towards the endplates under the action of surface tension. This final stage is impossible to observe in filament stretching experiments performed on earth as gravitational body forces are too great and cause sagging of the filament.

Images from the high definition video recordings have been used to obtain the mid-point filament diameter as a function of time during the stretching phase using a digital edge detection method implemented in Matlab. The data obtained from the video recording during stretching has been fitted with an exponential decay curve, and the time constant so

Task Progress:	obtained agrees with the reciprocal of the imposed extension rate. We use the diameter so obtained to calculate the extensional viscosity of the solution at different values of pre-shear rate and stretch rate. It is challenging to determine exact trends in extensional viscosity in the stretch regime due to lack of sensitivity of the force transducer. However, we are able to draw general conclusions and we determine the response of the test fluids to applied stretching deformations. We observe an increase in the extensional viscosity and Trouton ratio of around 2 orders of magnitude as the Hencky strain applied to the fluid increases. The magnitude of the extensional viscosity at high Hencky strains is comparable with that observed in ground-based CaBER experiments although it must be noted that the CaBER device does not allow control of an imposed extension rate. The limiting Trouton ratio for the viscoelastic suspension is nearly two orders of magnitude higher than the expected Trouton ratio for a Newtonian fluid in a uniaxial extensional flow, showing how the SHERE II test fluid is considerably strain hardening. We now examine the capillary thinning phase of the experiment following cessation of stretch. Analysis of the transient filament radius as a function of time reveals that the mid-plane diameter of the fluid column as it thins under the action of capillarity is well described by a single exponential decay. The time constant of this decay provides a measure of the fluid relaxation time, which is in good agreement with the results of ground-based small angle oscillatory shear (SAOS) rheometric testing. Upon calculating the extensional viscosity in the capillary thinning regime, we find that it is independent of the pre-shear rate imposed at the start of the experiment. However, we observe that the extensional viscosity increases with increasing strain rate imposed in the stretch regime. Thus we observe that the capillary thinning of the filament is dependent on the initial state of the filament when thinning begins. This initial state is influenced by the strain history of the filament imposed during the stretch phase. The SHERE and SHERE II experiments have allowed us for the first time to obtain accurate measurements of the extensional viscosity for less elastic fluids such as polymer solutions, suspensions, and liquid crystalline materials without sagging of the fluid under a gravitational body force. The new SHERE II data are of good quality, and further detailed examination utilizing the high-quality video data are expected to provide verification of existing theories for the effects of particles on the viscous behavior of complex non-Newtonian fluids. Understanding and quantifying the extensional rheology of complex fluids allows to improve and control earth-based manufacturing processes such as fiber micro/nanofabrication, atomization, electrospinning and to understand the flow and quantify the rheological properties of biofluids. The ultimate goal of SHERE II is to compare the differences in rheology between the filled polymer-particle system and the unfilled polymer system used during SHERE experiments.
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